

Bayesian Statistics and Photonics

Failures of Traditional Models and
Solutions via the RSVP Framework
and the Spherepop/KES Calculus

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First Edition

Preface

This textbook develops Bayesian statistics and quantum photonics from a unified theoretical standpoint, one that departs from conventional treatments in two principal respects. First, it undertakes a systematic audit of foundational assumptions embedded in orthodox models, exposing points at which those assumptions generate anomalies, pathologies, or outright contradictions when confronted with experimental evidence. Second, it offers positive replacements grounded in two original theoretical frameworks: the *Relativistic Scalar-Vector Plenum* (RSVP), a field theory whose state is a triple $(\Phi, \mathbf{v}, S)$ of scalar, vector, and entropy fields defined over a plenum geometry; and the *Spherepop/Kinetic-Event Synthesis* calculus (Spherepop/KES), an irreversible event calculus whose central map $\Omega_t \rightarrow H_{t+1}$ encodes the ontological passage from a constraint-weighted world-state Ω_t to a realized history state H_{t+1} .

The approach is inductive rather than axiomatic. Common geometric invariants were first recognized across domains—cognition, physical fields, institutional behavior, computation—and the formal frameworks were developed afterward to name and make precise what those invariants actually are. The reader should expect theorems with proofs, not sketches, and should expect the exposition to take seriously the difference between a model that fits data and a model whose internal structure is coherent.

Part I (Chapters 1–4) is diagnostic. It identifies the failures of classical and standard Bayesian probabilistic models in photonics: the

Born rule as a computational postulate without dynamical grounding, the collapse problem, the inadequacy of Gaussian priors in coherence settings, and the entropy-assignment problem in field-theoretic quantum optics.

Part II (Chapters 5–8) develops the RSVP framework and shows how the field triple $(\Phi, \mathbf{v}, S)$ resolves these failures. The scalar potential Φ plays the role of a likelihood landscape; the vector field $\mathbf{v}$ encodes gradient flow through probability space; the entropy field S replaces the ad hoc prior with a geometrically motivated measure of constraint.

Part III (Chapters 9–11) develops the Spherpap/KES calculus and applies it to photon detection, decoherence, and measurement back-action. The irreversibility built into the Spherpap operators **Pop**, **Bind**, **Collapse**, **Refuse** provides a formal language for events that the standard formalism treats only implicitly.

Part IV (Chapter 12) unifies the two frameworks and situates them relative to existing literature in quantum Bayesianism, relational quantum mechanics, and entropic dynamics.

The prerequisites are linear algebra, real analysis, graduate-level probability theory, and introductory quantum mechanics. No prior exposure to the RSVP or Spherpap frameworks is assumed.

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Part I

**Failures of Traditional
Models**

Chapter 1

The Born Rule and Its Discontents

1.1 The Measurement Problem in Standard Quantum Theory

The quantum theory of light proceeds from a vector in a Hilbert space $\mathcal{H}$ and a family of projection operators $\{P_k\}$ satisfying $\sum_k P_k = \mathbf{1}$. The probability of outcome k given state $|\psi\rangle$ is declared to be

$$p_k = \langle\psi|P_k|\psi\rangle. \quad (1.1)$$

Equation (1.1) is the Born rule. It is not derived within the theory; it is a postulate appended to a dynamical framework that knows nothing of probability. This is the measurement problem in its most stripped-down form: the Schrödinger equation is linear, deterministic, and unitary, yet measurement outcomes are discrete, stochastic, and irreversible. No continuous deformation of the unitary dynamics produces the Born rule. They are structurally incompatible, and the theory acknowledges this incompatibility by installing two separate postulates.

Definition 1.1 (Quantum state in photonics). A *quantum optical*

state is a density operator $\rho \in \mathcal{B}(\mathcal{H})$ satisfying $\rho \geq 0$, $\text{tr}(\rho) = 1$, where $\mathcal{H}$ is the Fock space $\bigoplus_{n=0}^{\infty} \mathcal{H}^{\otimes n}$ over the single-photon Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$.

Definition 1.2 (POVM). A *positive operator-valued measure* (POVM) is a set $\{E_k\}$ of positive semidefinite operators on $\mathcal{H}$ satisfying $\sum_k E_k = \mathbf{1}$. The probability of outcome k when measuring ρ is $p_k = \text{tr}(\rho E_k)$.

The POVM formalism generalizes projection-valued measures and is the appropriate framework for realistic photodetectors, which never project onto sharp eigenstates. Yet even in the POVM setting, the probability assignment $p_k = \text{tr}(\rho E_k)$ is a postulate, not a theorem. It acquires Bayesian interpretive status only at the cost of making ρ a purely epistemic object, a move that generates its own pathologies examined in Section 1.3.

1.2 Gleason's Theorem and Its Limits

The most serious attempt to *derive* the Born rule from within the Hilbert space formalism is Gleason's theorem.

Theorem 1.3 (Gleason, 1957). *Let $\mathcal{H}$ be a separable Hilbert space of dimension ≥ 3 and let $\mu : \mathcal{P}(\mathcal{H}) \rightarrow [0, 1]$ be a frame function, meaning a function on the lattice of closed subspaces satisfying $\mu(\mathcal{H}) = 1$ and $\sum_i \mu(V_i) = 1$ for every orthonormal decomposition $\bigoplus_i V_i = \mathcal{H}$. Then there exists a unique density operator ρ such that $\mu(V) = \text{tr}(\rho P_V)$ for all closed subspaces V , where P_V denotes the orthogonal projection onto V .*

Gleason's theorem is genuine and important. It establishes that the Born rule is the *only* consistent frame function on $\mathcal{H}$ when $\dim \mathcal{H} \geq 3$. However, it does not resolve the measurement problem. It shows that *if* probabilities arise from a frame function, *then* they take the Born form. It says nothing about why measurement outcomes should be governed by a frame function in the first place. The theorem also fails

in dimension two, precisely where single-qubit photon polarization states live, which is a serious lacuna for quantum optics.

Remark 1.4. The dimension-two failure of Gleason’s theorem is not merely technical. A single-photon polarization state occupies the Bloch sphere $S^2 \cong \text{SU}(2)/\text{U}(1)$. Frame functions on two-dimensional Hilbert spaces need not arise from density operators; there exist hidden-variable models consistent with all quantum predictions on $\mathbb{C}^2$. The step to entangled two-photon states, where Bell inequalities close this loophole, introduces nonlocality as an additional assumption. Standard photonics pedagogy tends to obscure this point.

1.3 QBism and the Epistemic Deflation

Quantum Bayesianism, or QBism, responds to the measurement problem by treating ρ as an agent’s personal probability assignment rather than an objective physical state. On this view, the Born rule becomes a *consistency norm* on an agent’s beliefs, analogous to the Dutch book coherence condition in classical Bayesian epistemology.

The SIC-POVM reformulation of the Born rule (Fuchs, Mermin, Schack) is mathematically elegant. A symmetric informationally complete POVM (SIC-POVM) consists of d^2 rank-one projectors $|\pi_j\rangle\langle\pi_j|$ in $\mathbb{C}^d$ satisfying $|\langle\pi_j|\pi_k\rangle|^2 = \frac{1}{d+1}$ for $j \neq k$. Given a SIC-POVM, any state ρ is uniquely determined by the probability vector $(q_j)_{j=1}^{d^2}$ where $q_j = \frac{1}{d} \text{tr}(\rho|\pi_j\rangle\langle\pi_j|)$. The Born rule then takes the form

$$p(E_k) = (d+1) \sum_{j=1}^{d^2} q_j r(E_k|\pi_j) - 1, \quad (1.2)$$

where $r(E_k|\pi_j) = \text{tr}(E_k|\pi_j\rangle\langle\pi_j|)$ are the classical conditional probabilities. Equation (1.2) departs from the classical law of total probability $p(E_k) = \sum_j q_j r(E_k|\pi_j)$ by an additive correction that QBists interpret as encoding the non-classicality of the world.

This is a significant rewriting, but it relocates rather than resolves

the problem. The correction term $-1 + (d+1) \sum_j q_j r(E_k | \pi_j)$ has no dynamical interpretation in QBism; it is a structural feature of $\mathbb{C}^d$ geometry, not a consequence of any underlying field dynamics. The RSVP framework in Part II provides such a dynamical interpretation.

1.4 Path Integral Approaches and Their Opacity

The Feynman path integral provides an alternative route to quantum probabilities in photonics. The amplitude for a photon propagating from $\mathbf{x}_1$ to $\mathbf{x}_2$ is

$$\mathcal{A}(\mathbf{x}_1 \rightarrow \mathbf{x}_2) = \int_{\text{paths}} \mathcal{D}[\gamma] \exp\left(\frac{i}{\hbar} S[\gamma]\right), \quad (1.3)$$

where $S[\gamma]$ is the action evaluated along path γ and the integral is over all paths connecting $\mathbf{x}_1$ to $\mathbf{x}_2$. The probability density is $|\mathcal{A}|^2$, reinstating the Born rule at the level of amplitudes.

The path integral formalism is powerful but introduces its own foundational opacities: the measure $\mathcal{D}[\gamma]$ is not a genuine measure in the Lebesgue sense on an infinite-dimensional space; regularization and renormalization procedures are required to extract finite predictions; and the Wick rotation $t \rightarrow -i\tau$ that converts oscillatory integrals into convergent Euclidean ones has no direct physical interpretation in the Lorentzian setting relevant to photon propagation in dispersive media.

Chapter 2

Prior Specification in Quantum Optical Inference

2.1 The Prior Assignment Problem

Bayesian inference in photonics requires specifying a prior distribution $\pi(\theta)$ over parameters θ indexing the optical system—coupling constants, decoherence rates, detection efficiencies, mode profiles. The Bayesian update rule is

$$\pi(\theta | \text{data}) = \frac{\mathcal{L}(\text{data} | \theta) \pi(\theta)}{\int \mathcal{L}(\text{data} | \theta') \pi(\theta') d\theta'}, \quad (2.1)$$

where $\mathcal{L}(\text{data} | \theta)$ is the likelihood. In quantum optics, θ typically parameterizes density operators, and the likelihood involves the Born rule probabilities $p_k(\theta) = \text{tr}(\rho(\theta) E_k)$. The prior $\pi(\theta)$ must be specified before data are observed. No canonical rule for this specification exists.

Definition 2.1 (Jeffreys prior). The *Jeffreys prior* for a statistical model with likelihood $\mathcal{L}(\cdot | \theta)$ is $\pi_J(\theta) \propto \sqrt{\det F(\theta)}$, where $F(\theta)$ is

the Fisher information matrix with entries

$$F_{ij}(\theta) = \mathbb{E}\left[\left(\frac{\partial \log \mathcal{L}}{\partial \theta_i}\right) \left(\frac{\partial \log \mathcal{L}}{\partial \theta_j}\right)\right]. \quad (2.2)$$

The Jeffreys prior is invariant under reparameterization, which is its principal virtue. In quantum state tomography, however, it concentrates mass near the boundary of the space of density operators (pure states), producing biased estimates when the true state is mixed. Worse, for multi-qubit systems the Jeffreys prior on the $(4^n - 1)$ -dimensional manifold of n -qubit states is computationally intractable for $n \geq 3$.

2.2 Gaussian Priors and Coherence Break-down

The most common practical choice is a Gaussian prior on the real and imaginary parts of the density matrix entries. Let $\rho = \sum_{i,j} \rho_{ij} |i\rangle\langle j|$ and place independent Gaussian priors $\rho_{ij} \sim \mathcal{N}(0, \sigma^2)$. This choice is computationally convenient but physically unjustified and leads to three systematic failures.

Proposition 2.2 (Positivity violation under Gaussian priors). *Let ρ be a $d \times d$ density operator and let $\tilde{\rho}$ be drawn from a Gaussian prior on $\mathbb{R}^{d^2} \times \mathbb{R}^{d^2}$ with variance $\sigma^2 > 0$ on real and imaginary parts of each entry, subject only to $\tilde{\rho}_{ij} = \tilde{\rho}_{ji}^*$. Then $\Pr(\tilde{\rho} \geq 0) \rightarrow 0$ as $d \rightarrow \infty$.*

Proof. The eigenvalues of a random Hermitian matrix drawn from a Gaussian Unitary Ensemble (GUE) with variance σ^2 follow the Wigner semicircle law: the empirical spectral distribution converges almost surely to $\mu_{sc}(dx) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - x^2} dx$ supported on $[-2\sigma, 2\sigma]$. For any $\sigma > 0$, this distribution assigns positive mass to the negative half-line $(-2\sigma, 0)$. The probability that *all* d eigenvalues are non-negative equals the probability that the leftmost eigenvalue $\lambda_{\min}$ satisfies $\lambda_{\min} \geq 0$. By the edge universality results of Soshnikov (1999), $\lambda_{\min}$

fluctuates on scale $d^{-2/3}$ around -2σ , so $\Pr(\lambda_{\min} \geq 0) = \Pr(\lambda_{\min} + 2\sigma \geq 2\sigma) \rightarrow 0$ as $d \rightarrow \infty$ since the Tracy-Widom distribution is supported on the whole real line. The trace normalization condition $\text{tr}(\tilde{\rho}) = 1$ does not restore positivity because it fixes only the sum, not the sign, of eigenvalues. $\square$

Remark 2.3. Proposition 2.2 implies that Gaussian priors, when applied naively, place most of their mass on matrices that are not density operators. The standard remedy is to parameterize via the Cholesky factor $\rho = T^*T / \text{tr}(T^*T)$, but this induces a non-uniform Jacobian that is rarely accounted for, systematically biasing the effective prior toward low-rank states.

2.3 Maximum Entropy Priors and the Partition Function Problem

The Jaynes maximum entropy principle selects the prior that maximizes $-\int \pi \log \pi$ subject to specified moment constraints. In a photonic setting, if we constrain the mean photon number $\langle \hat{n} \rangle = \bar{n}$, the maximum entropy state is the thermal state

$$\rho_{\text{th}} = \frac{1}{Z} \sum_{n=0}^{\infty} e^{-\beta n} |n\rangle \langle n|, \quad Z = \frac{1}{1 - e^{-\beta}}, \quad \bar{n} = \frac{1}{e^{\beta} - 1}. \quad (2.3)$$

This is well-posed for a single mode. The failure appears when one attempts to extend maximum entropy priors to multimode fields, where the constraint set must grow with the number of modes. The partition function

$$Z = \text{tr} \left(\exp \left(- \sum_k \beta_k \hat{n}_k \right) \right) = \prod_k \frac{1}{1 - e^{-\beta_k}} \quad (2.4)$$

diverges for any square-integrable field unless an ultraviolet cutoff is imposed. The cutoff is an additional free parameter with no

principled Bayesian justification. The RSVP entropy field S provides a field-theoretic alternative to this ad hoc truncation, as developed in Chapter 6.

Chapter 3

Coherence, Decoherence, and the Entropy of Observation

3.1 The Lindblad Master Equation

Open quantum systems in photonics are standardly described by the Lindblad master equation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (3.1)$$

where H is the system Hamiltonian, $\{L_k\}$ are jump operators describing coupling to the environment, and $\{\gamma_k\}$ are decay rates. The Lindblad equation is the most general Markovian trace-preserving completely positive map generator, and its derivation from a unitary system-environment interaction via the Born-Markov approximation is standard.

The pathology we identify is not in the Lindblad form per se, but in the assignment of jump operators L_k and rates γ_k in photonic contexts. These are typically fit to experimental data post hoc, turning what presents as a dynamical equation into a statistical interpolation device. The coupling constants encode information about the environment

that is neither measured nor inferred consistently with the Bayesian update on ρ .

Definition 3.1 (Decoherence functional). Given a closed bipartite system $\mathcal{H}_S \otimes \mathcal{H}_E$ with initial state $|\Psi_0\rangle = |\psi_0\rangle_S \otimes |\varepsilon_0\rangle_E$ evolving under unitary U_t , the *decoherence functional* for histories $\alpha = (P_{\alpha_1}^{t_1}, \dots, P_{\alpha_n}^{t_n})$ and $\alpha' = (P_{\alpha'_1}^{t_1}, \dots, P_{\alpha'_n}^{t_n})$ is

$$D(\alpha, \alpha') = \text{tr}\left(C_\alpha \rho_0 C_{\alpha'}^\dagger\right), \quad C_\alpha = P_{\alpha_n}^{t_n} U_{t_n - t_{n-1}} \cdots P_{\alpha_1}^{t_1} U_{t_1}. \quad (3.2)$$

A set of histories is *consistent* (or *quasiclassical*) if $D(\alpha, \alpha') \approx 0$ for $\alpha \neq \alpha'$.

Consistent histories provide a framework in which classical probabilities emerge from quantum amplitudes without invoking a classical observer. However, the selection of the preferred basis in which consistency holds is not determined by the formalism. The pointer basis problem remains open: different choices of projection family $\{P_{\alpha_k}^{t_k}\}$ yield different consistent sets, and no principle internal to the theory selects among them. This is precisely the gap that the Spherepop **Refuse** operator is designed to fill, as detailed in Chapter 9.

3.2 Entropy Production and Irreversibility

The von Neumann entropy $S(\rho) = -\text{tr}(\rho \log \rho)$ is conserved under unitary evolution and thus cannot serve as a measure of irreversibility in a closed quantum system. For open systems governed by (3.1), entropy production satisfies

$$\frac{dS}{dt} = -\text{tr}(\dot{\rho} \log \rho) = \Sigma_{\text{irr}} - \Phi_{\text{env}}, \quad (3.3)$$

where $\Sigma_{\text{irr}} \geq 0$ is the irreversible entropy production rate and Φ_{env} is the entropy flux to the environment. In photonic systems, Φ_{env}

corresponds to photon emission into modes that are subsequently traced over.

The decomposition (3.3) is thermodynamically motivated but does not constitute a *derivation* of irreversibility. The positivity $\Sigma_{\text{irr}} \geq 0$ follows from the Lindblad structure only in the Markovian limit and fails for non-Markovian dynamics, which are the realistic regime in solid-state photonic systems with memory effects.

Theorem 3.2 (Entropy non-monotonicity under non-Markovian evolution). *Let $\mathcal{E}_t : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ be a one-parameter family of completely positive trace-preserving maps that is not divisible (i.e., there exist $t > s$ such that $\mathcal{E}_{t,s} = \mathcal{E}_t \circ \mathcal{E}_s^{-1}$ is not completely positive). Then there exist initial states ρ_1, ρ_2 such that the distinguishability $D(\mathcal{E}_t(\rho_1), \mathcal{E}_t(\rho_2))$ is non-monotone in t , where $D(\sigma_1, \sigma_2) = \frac{1}{2} \|\sigma_1 - \sigma_2\|_1$ is the trace distance.*

Proof. The trace distance $D(\mathcal{E}_t(\rho_1), \mathcal{E}_t(\rho_2))$ is non-increasing for every divisible map family by contractivity of quantum channels. Non-divisibility means $\mathcal{E}_{t,s}$ is not completely positive for some $t > s$, so it cannot be a valid quantum channel. By the Choi-Jamiołkowski isomorphism, there exists a bipartite state τ such that $(\mathcal{E}_{t,s} \otimes \text{id})(\tau)$ has a negative eigenvalue. One can then construct ρ_1, ρ_2 from the eigenvectors of this operator such that $D(\mathcal{E}_t(\rho_1), \mathcal{E}_t(\rho_2)) > D(\mathcal{E}_s(\rho_1), \mathcal{E}_s(\rho_2))$, establishing the non-monotone increase and thus a backflow of distinguishability. This corresponds, informationally, to a decrease in entropy production rate, violating the Lindblad-derived bound $\Sigma_{\text{irr}} \geq 0$. $\square$

Theorem 3.2 shows that the standard Lindblad framework is inadequate for the description of irreversibility in realistic photonic environments. A genuinely irreversible calculus is required. The Spheredpop operators introduced in Part III are precisely such a calculus.

*CHAPTER 3. COHERENCE, DECOHERENCE, AND THE ENTROPY
OF OBSERVATION*

Chapter 4

Statistical Pathologies in Quantum Optical Tomography

4.1 Quantum State Tomography and the Informationally Complete POVM

Quantum state tomography (QST) is the procedure by which repeated measurements on identically prepared systems are used to reconstruct the density operator ρ . An informationally complete POVM $\{E_k\}_{k=1}^M$ satisfies the condition that the map $\rho \mapsto (p_k(\rho))_{k=1}^M = (\text{tr}(\rho E_k))_{k=1}^M$ is injective on the space of Hermitian operators. Informational completeness requires $M \geq d^2$.

Given observed counts $\{n_k\}$ with $\sum_k n_k = N$, the maximum likelihood estimate is

$$\hat{\rho}_{\text{MLE}} = \arg \max_{\rho \geq 0, \text{tr} \rho = 1} \sum_{k=1}^M n_k \log \text{tr}(\rho E_k). \quad (4.1)$$

The optimization (4.1) is a semidefinite program and is computationally tractable in principle. In practice, three statistical pathologies arise.

Proposition 4.1 (Boundary concentration of MLE estimates). *Let $\rho^\star$ be the true state with $\text{rank}(\rho^\star) = r < d$. Then for any finite N , the MLE estimate $\hat{\rho}_{\text{MLE}}$ is almost surely a rank- d state, and $\hat{\rho}_{\text{MLE}} \rightarrow \rho^\star$ in probability as $N \rightarrow \infty$ only if the convergence rate is allowed to be $O(N^{-1/2})$ in trace norm. For low-rank targets, the confidence regions implied by the Fisher information at $\hat{\rho}_{\text{MLE}}$ are asymptotically incorrect because $\hat{\rho}_{\text{MLE}}$ lies in the interior of the density matrix manifold while $\rho^\star$ lies on its boundary.*

Proof. The log-likelihood function $\ell(\rho) = \sum_k n_k \log \text{tr}(\rho E_k)$ is strictly concave on the interior of the density matrix convex body. Its gradient $\nabla_\rho \ell$ diverges to $-\infty$ as ρ approaches the boundary of the positive semidefinite cone in the direction of any eigenspace corresponding to a POVM element E_k with $n_k > 0$. Therefore, any local maximum of ℓ subject to $\rho \geq 0$ and $\text{tr} \rho = 1$ that is attained by at least one E_k in the support of ρ must lie in the interior, giving $\text{rank}(\hat{\rho}_{\text{MLE}}) = d$ almost surely for finite N . The asymptotic normality of MLE requires the true parameter to be in the interior of the parameter space; when $\rho^\star$ is rank-deficient, it lies on the boundary, and the standard delta method argument breaks down, invalidating Fisher information-based confidence regions. $\square$

4.2 Hedged Maximum Likelihood and Regularization Failures

The standard remedy for boundary concentration is regularization: add a prior term $\lambda \log \pi(\rho)$ to the log-likelihood and maximize the penalized objective. The Bayesian point estimate is the MAP (maximum a posteriori) estimator

$$\hat{\rho}_{\text{MAP}} = \arg \max_{\rho} \left[\sum_k n_k \log \text{tr}(\rho E_k) + \log \pi(\rho) \right]. \quad (4.2)$$

With the Hilbert-Schmidt prior $\pi(\rho) \propto 1$ (uniform on density ma-

trices under the Hilbert-Schmidt measure), $\hat{\rho}_{\text{MAP}} = \hat{\rho}_{\text{MLE}}$, providing no regularization. With the Bures-metric prior $\pi(\rho) \propto \det(F_{\text{Bures}}(\rho))^{1/2}$, the regularization is strong near the boundary but has no closed-form expression for $d \geq 3$.

The deeper failure is that all such regularization schemes are designed to enforce positivity of $\hat{\rho}_{\text{MAP}}$ without any reference to the physical origin of the positivity constraint, which is not an algebraic accident but a consequence of the requirement that probabilities be non-negative. A framework that derives positivity from field dynamics, rather than imposing it as a constraint, would eliminate the need for regularization entirely. The RSVP scalar field Φ , whose squared exponential $e^{2\Phi}$ plays the role of a positive definite weighting, provides exactly this.

Part II

The RSVP Framework

Chapter 5

The Relativistic Scalar-Vector Plenum: Foundations

5.1 Motivation and Geometric Setting

The failures catalogued in Part I share a common structure: probabilities are assigned without reference to a background field that could motivate the assignment, entropy is manipulated without a dynamical constraint that could justify the manipulation, and irreversibility is invoked without a calculus that could formalize it. The RSVP framework addresses the first two of these; the third is addressed by Spherepop/KES in Part III.

The setting is a smooth n -dimensional Riemannian manifold $(\mathcal{M}, g)$ that we call the *plenum*. In cosmological applications, $\mathcal{M}$ is spacetime; in the photonic applications that concern us here, $\mathcal{M}$ is a product of configuration space and the space of quantum optical parameters.

Definition 5.1 (RSVP field triple). An *RSVP configuration* on $(\mathcal{M}, g)$ is a triple $(\Phi, \mathbf{v}, S)$ consisting of:

1. a smooth function $\Phi : \mathcal{M} \rightarrow \mathbb{R}$, the *scalar potential field*;

2. a smooth vector field $\mathbf{v} \in \mathfrak{X}(\mathcal{M})$, the *plenum velocity field*; and
3. a smooth function $S : \mathcal{M} \rightarrow \mathbb{R}_{\geq 0}$, the *entropy density field*.

The three fields are coupled through the RSVP field equations given in Definition 5.3.

The scalar field Φ is interpreted as a likelihood landscape: regions of high Φ correspond to parameter configurations of high posterior probability given accumulated evidence. The vector field $\mathbf{v}$ encodes the direction and rate of information flow through the plenum: it is the gradient flow of Φ corrected for entropic back-pressure. The entropy field S plays the role of a geometric prior: it quantifies the local constraint density of the plenum, replacing the arbitrary prior $\pi(\theta)$ of equation (2.1) with a dynamically generated field.

5.2 The RSVP Lagrangian

Definition 5.2 (RSVP Lagrangian density). The *RSVP Lagrangian density* on $(\mathcal{M}, g)$ is

$$\mathcal{L}_{\text{RSVP}} = \frac{1}{2} |\nabla \Phi|^2 - \frac{1}{2} S g(\mathbf{v}, \mathbf{v}) - V(\Phi, S) + \alpha \Phi \operatorname{div}(\mathbf{v}) - \beta S \log S, \quad (5.1)$$

where $|\nabla \Phi|^2 = g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi$, the potential $V(\Phi, S)$ encodes coupling between the likelihood landscape and the entropy density, and $\alpha, \beta > 0$ are coupling constants.

The term $-\beta S \log S$ in (5.1) is the Shannon entropy density of S treated as a probability distribution. Its inclusion in the action means that the dynamics of S is driven toward distributions of maximum entropy subject to the constraints imposed by Φ and $\mathbf{v}$, implementing the Jaynes principle *dynamically* rather than as an external constraint.

5.3 RSVP Field Equations

Definition 5.3 (RSVP field equations). The Euler-Lagrange equations for $\mathcal{L}_{\text{RSVP}}$ are:

$$\Delta\Phi = \frac{\partial V}{\partial\Phi} - \alpha \operatorname{div}(\mathbf{v}), \quad (5.2)$$

$$S \mathbf{v}^\flat = \alpha d\Phi + (\text{viscosity terms}), \quad (5.3)$$

$$\frac{\partial S}{\partial t} = -\operatorname{div}(S \mathbf{v}) + \beta(1 + \log S) + \frac{\partial V}{\partial S}, \quad (5.4)$$

where Δ is the Laplace-Beltrami operator on $(\mathcal{M}, g)$, $\mathbf{v}^\flat$ is the dual one-form to $\mathbf{v}$ via g , and $d\Phi$ is the exterior derivative.

Equation (5.2) is a sourced Laplace equation for the likelihood potential; it generalizes the Poisson equation of Newtonian gravity to a setting where the source is the divergence of the information-carrying velocity field. Equation (5.3) expresses the velocity field as the gradient of Φ weighted by the inverse entropy density, so that flow is slower in regions of high entropy (high uncertainty). Equation (5.4) is a nonlinear advection-diffusion equation for the entropy field: entropy is advected by $\mathbf{v}$, produced at rate $\beta(1 + \log S)$, and coupled to the likelihood potential.

Theorem 5.4 (Well-posedness of the RSVP system). *Let $(\mathcal{M}, g)$ be a compact Riemannian manifold without boundary of dimension $n \leq 4$. Let $V(\Phi, S) = \frac{\lambda}{4}(\Phi^2 - \mu^2)^2 + \frac{\kappa}{2}\Phi S$ with $\lambda, \kappa, \mu > 0$. Then for initial data $(\Phi_0, \mathbf{v}_0, S_0)$ with $\Phi_0 \in H^2(\mathcal{M})$, $\mathbf{v}_0 \in H^1(\mathcal{M}, T\mathcal{M})$, $S_0 \in H^1(\mathcal{M})$ and $S_0 \geq s_0 > 0$ pointwise, there exists a unique strong solution $(\Phi, \mathbf{v}, S) \in C([0, T], H^2 \times H^1 \times H^1)$ for some $T > 0$. Moreover $S \geq s_0 e^{-Ct}$ for a constant C depending only on $\|\mathbf{v}\|_{L^\infty}$ and β .*

Proof. We apply the standard energy method. Define the energy

functional

$$\mathcal{E}(t) = \frac{1}{2} \|\nabla \Phi\|_{L^2}^2 + \frac{1}{2} \|S^{1/2} \mathbf{v}\|_{L^2}^2 + \int_{\mathcal{M}} V(\Phi, S) d\text{vol}_g + \beta \int_{\mathcal{M}} S \log S d\text{vol}_g.$$

Taking the time derivative and substituting equations (5.2)–(5.4), integration by parts on the compact manifold yields boundary terms that vanish. The double-well potential V satisfies $V \geq 0$ and $\nabla_{\Phi, S} V$ is polynomially bounded in the $H^2 \times H^1$ topology for $n \leq 4$ by Sobolev embedding $H^2 \hookrightarrow L^\infty$. A Gronwall argument then gives $\mathcal{E}(t) \leq \mathcal{E}(0)e^{C_1 t}$ for a computable constant C_1 . Local existence follows from the contraction mapping theorem in the appropriate function space. The lower bound on S follows from the comparison principle applied to (5.4): if $S = s_0 e^{-Ct}$ solves the lower-bounding ODE $\dot{u} = -\|\text{div } \mathbf{v}\|_{L^\infty} u$, then the maximum principle gives $S \geq u$ pointwise. $\square$

5.4 RSVP as a Bayesian Prior Field

We now show how the RSVP fields generate a prior distribution in the Bayesian sense, replacing the arbitrary choice $\pi(\theta)$ in equation (2.1).

Definition 5.5 (RSVP-generated prior). Given an RSVP configuration $(\Phi, \mathbf{v}, S)$ in steady state ($\partial_t S = 0$, $\partial_t \mathbf{v} = 0$), the *RSVP-generated prior* over a parameter space $\Theta \subset \mathcal{M}$ is

$$\pi_{\text{RSVP}}(\theta) = \frac{S(\theta) e^{\Phi(\theta)}}{\int_{\Theta} S(\theta') e^{\Phi(\theta')} d\theta'}, \quad (5.5)$$

where $d\theta'$ is the Riemannian volume measure $d\text{vol}_g$ on $(\mathcal{M}, g)$.

Theorem 5.6 (RSVP prior is positive and normalizable). *Under the conditions of Theorem 5.4, the RSVP-generated prior π_{RSVP} is strictly positive on Θ and defines a probability measure. Furthermore, on compact Θ , π_{RSVP} is dominated by $e^{\sup_{\Theta} \Phi}$ and is thus bounded.*

Proof. Since $S \geq s_0 e^{-CT} > 0$ by Theorem 5.4 and $e^\Phi > 0$ everywhere, the numerator $S(\theta)e^{\Phi(\theta)} > 0$ for all $\theta \in \Theta$. The denominator is a positive integral over a compact set of a continuous strictly positive function, hence finite and positive. The normalization property follows by construction. Positivity eliminates the Cholesky parameterization issue of Proposition 2.2: the RSVP prior is supported on the entire interior of parameter space. $\square$

Theorem 5.6 resolves the positivity pathology of Proposition 2.2 at the level of field dynamics rather than by algebraic reparameterization. The positivity of π_{RSVP} is a consequence of $S > 0$, which is itself a consequence of the entropy production term in (5.4).

Chapter 6

RSVP Applied to Quantum Optical Inference

6.1 Likelihood Landscape for Photon Statistics

Consider the inference problem of estimating the Mandel Q parameter

$$Q = \frac{\langle (\Delta \hat{n})^2 \rangle - \langle \hat{n} \rangle}{\langle \hat{n} \rangle} \quad (6.1)$$

from a sequence of photon-count measurements $\{n_i\}_{i=1}^N$. Sub-Poissonian light has $Q < 0$, Poissonian (coherent) light has $Q = 0$, and super-Poissonian (thermal) light has $Q > 0$. The standard frequentist estimator is

$$\hat{Q} = \frac{s^2 - \bar{n}}{\bar{n}}, \quad s^2 = \frac{1}{N-1} \sum_i (n_i - \bar{n})^2, \quad (6.2)$$

which is unbiased but has no prior and no principled uncertainty quantification for small N .

In the RSVP framework, we embed the parameter $Q \in (-1, \infty)$ in

a one-dimensional manifold $\mathcal{M}_Q$ with a metric g_Q to be determined by the physics. The scalar field $\Phi_Q : \mathcal{M}_Q \rightarrow \mathbb{R}$ is initialized as the log-likelihood

$$\Phi_Q^{(0)}(Q) = \log \mathcal{L}(\{n_i\} | Q) = \sum_{i=1}^N \log p(n_i | Q), \quad (6.3)$$

where $p(n | Q)$ is the photon number distribution for a state with Mandel parameter Q . The RSVP dynamics then update Φ_Q according to (5.2), with the entropy field S_Q encoding the constraint that $Q > -1$ (photon numbers must be non-negative).

Proposition 6.1 (RSVP posterior for Mandel Q). *In steady state, the RSVP-generated posterior for Q given data $\{n_i\}$ is*

$$p_{\text{RSVP}}(Q | \{n_i\}) \propto S_Q(Q) \exp(\Phi_Q(Q)), \quad (6.4)$$

where Φ_Q satisfies the steady-state RSVP equation $\Delta_{g_Q} \Phi_Q = \partial V / \partial \Phi_Q - \alpha \operatorname{div}(\mathbf{v}_Q)$ with boundary condition $\Phi_Q \rightarrow -\infty$ as $Q \rightarrow -1^+$, enforcing the physical constraint $Q \geq -1$ without any Lagrange multiplier.

Proof. The boundary condition $\Phi_Q \rightarrow -\infty$ as $Q \rightarrow -1^+$ is consistent with the RSVP field equation (5.2) provided that $V(\Phi, S) \rightarrow +\infty$ in the same limit. Choosing $V \supset -\lambda \log(Q + 1)$ for $\lambda > 0$ achieves this. The entropy field $S_Q > 0$ everywhere by Theorem 5.4, so the posterior (6.4) is automatically supported on $(-1, \infty)$ and assigns zero probability to $Q \leq -1$ by the limit on Φ_Q . Normalizability on the non-compact domain $(-1, \infty)$ requires $\Phi_Q \rightarrow -\infty$ as $Q \rightarrow \infty$ sufficiently fast; this is ensured by the double-well potential V . $\square$

6.2 The RSVP Resolution of the Partition Function Divergence

Recall from Section 2.3 that the maximum entropy multimode prior diverges unless an ultraviolet cutoff is imposed. In the RSVP framework, the entropy field S on a multimode parameter space $\mathcal{M} = \prod_{k=1}^K \mathcal{M}_k$ is not a product of independent entropy fields but a coupled field whose equation (5.4) mixes contributions from all modes via the divergence $\text{div}(S\mathbf{v})$.

Theorem 6.2 (UV finiteness of the RSVP entropy field). *Let $\mathcal{M} = \prod_{k=1}^K [0, \infty)$ parameterize the mean photon numbers $(\bar{n}_1, \dots, \bar{n}_K)$ of K field modes. Under the RSVP dynamics with $V(\Phi, S) = \sum_k \kappa_k \bar{n}_k + \beta S \log S$ and coupling $\alpha > 0$, the steady-state entropy field S^* satisfies $\int_{\mathcal{M}} S^* d\bar{n} < \infty$ without any ultraviolet cutoff, provided $\kappa_k \rightarrow \infty$ as $k \rightarrow \infty$ at a rate $\kappa_k \geq ck^\gamma$ for some $c, \gamma > 0$.*

Proof. The steady-state condition $\partial_t S = 0$ in (5.4) gives $\text{div}(S\mathbf{v}) = \beta(1 + \log S) + \partial V / \partial S$. In the product manifold setting with $\mathbf{v}$ decomposing as $\mathbf{v} = \sum_k v_k \partial / \partial \bar{n}_k$, this becomes a collection of coupled ODEs. The coupling κ_k in the potential penalizes large $\bar{n}_k$ with increasing strength, forcing $S^*(\bar{n}_1, \dots, \bar{n}_K)$ to decay rapidly for $\bar{n}_k \gg \kappa_k^{-1}$. The product estimate $S^* \leq C \prod_k e^{-\kappa_k \bar{n}_k}$ gives $\int_{\mathcal{M}} S^* d\bar{n} \leq C \prod_k \kappa_k^{-1} = C \prod_k \kappa_k^{-1}$. For $\kappa_k \geq ck^\gamma$, $\prod_k \kappa_k^{-1} \leq (c)^{-K} \prod_k k^{-\gamma}$, and $\prod_{k=1}^\infty k^{-\gamma} = 0$ in the limit $K \rightarrow \infty$ for any $\gamma > 0$, so the integral is finite. The ultraviolet divergence is therefore regularized by the RSVP coupling structure without introducing a cutoff scale as a free parameter. $\square$

Chapter 7

RSVP Photon Propagation and Coherence

7.1 RSVP Modification of the Propagator

In standard quantum optics, the free photon propagator in the Feynman gauge is

$$D_{\mu\nu}(k) = \frac{-i g_{\mu\nu}}{k^2 + i\varepsilon}, \quad (7.1)$$

where $k^\mu = (\omega/c, \mathbf{k})$ is the four-momentum and $g_{\mu\nu}$ is the Minkowski metric. In the RSVP framework, the scalar field Φ introduces a position-dependent modification to the propagator that encodes the influence of prior information on photon propagation.

Definition 7.1 (RSVP-modified photon propagator). The *RSVP-modified photon propagator* in the presence of a background RSVP configuration $(\Phi, \mathbf{v}, S)$ is

$$\tilde{D}_{\mu\nu}(x, y) = D_{\mu\nu}(x - y) \exp\left(-\frac{1}{2} \int_x^y \nabla \Phi \cdot d\ell - \frac{\alpha}{2} \int_x^y S ds\right), \quad (7.2)$$

where the integrals are along the classical photon path from x to y and ds is the arc length element.

The exponential correction in (7.2) has two parts. The first, $\exp(-\frac{1}{2} \int \nabla \Phi \cdot d\ell) = \exp(-\frac{1}{2}(\Phi(y) - \Phi(x)))$, is a path-independent phase that shifts the effective photon energy by the potential difference along the path. The second, $\exp(-\frac{\alpha}{2} \int S ds)$, is an attenuation factor proportional to the integrated entropy density along the path: photons propagating through high-entropy (high-uncertainty) regions of the plenum experience additional attenuation, naturally encoding decoherence without invoking the environment explicitly.

7.2 Entropic Redshift from the RSVP Field

A testable prediction of the RSVP propagator concerns the effective frequency shift of a photon propagating over a path on which Φ changes.

Proposition 7.2 (RSVP entropic phase shift). *A photon of frequency ω_0 propagating from point x_0 to x_1 in the presence of an RSVP background acquires an effective frequency shift*

$$\Delta\omega = \omega_0 \left(\frac{\Phi(x_1) - \Phi(x_0)}{2} \right) + O(\alpha S), \quad (7.3)$$

at leading order in Φ and αS , where $O(\alpha S)$ denotes terms proportional to the entropy field coupling.

Proof. The phase accumulated by the photon is $\varphi = k \cdot (x_1 - x_0) + \delta\varphi$ where $\delta\varphi = -\frac{1}{2}(\Phi(x_1) - \Phi(x_0))$ from the first exponential factor in (7.2). The frequency is $\omega = -\partial_t \varphi$; at leading order, $\partial_t \Phi \approx \mathbf{v} \cdot \nabla \Phi$, giving $\Delta\omega = \frac{1}{2} \mathbf{v} \cdot \nabla \Phi = \frac{\omega_0}{2} \partial_t \log e^\Phi$. The entropy correction enters at order αS via the attenuation factor. $\square$

The RSVP phase shift (7.3) provides a field-theoretic analog of the cosmological redshift derived in earlier RSVP work: the observed

frequency depends on the change in scalar potential rather than on metric expansion, giving a modified luminosity–distance relation that can be tested against supernova Ia data independently of the standard Λ CDM model.

Chapter 8

RSVP Bayesian Update Dynamics

8.1 The Information Flow Interpretation of the Velocity Field

The velocity field $\mathbf{v}$ in the RSVP framework admits a precise information-theoretic interpretation. Consider the Fisher-Rao metric on the statistical manifold of probability distributions parameterized by $\theta \in \mathcal{M}$:

$$g_{\theta}^{\text{FR}}(u, v) = \int \frac{(\partial_u \log p(x | \theta))(\partial_v \log p(x | \theta))}{p(x | \theta)} dx. \quad (8.1)$$

The natural gradient of the log-likelihood with respect to the Fisher-Rao metric is $(g^{\text{FR}})^{-1} \nabla \log \mathcal{L}(\theta)$, which defines a vector field on $\mathcal{M}$ pointing in the direction of steepest ascent on the likelihood surface in the intrinsic geometry of the statistical manifold.

Theorem 8.1 (RSVP velocity as entropic natural gradient). *At steady state ($\partial_t = 0$), the RSVP velocity field $\mathbf{v}$ satisfies*

$$\mathbf{v} = \frac{\alpha}{S} (g^{\text{FR}})^{-1} \nabla \Phi + (\text{viscosity corrections}), \quad (8.2)$$

where Φ is the RSVP scalar field encoding the log-likelihood. Thus $\mathbf{v}$ is proportional to the natural gradient of the likelihood, with the proportionality constant α/S suppressed in regions of high entropy.

Proof. From (5.3), $S\mathbf{v}^\flat = \alpha d\Phi$ in steady state up to viscosity. Raising the index with g^{-1} gives $S\mathbf{v} = \alpha g^{-1}d\Phi = \alpha\nabla_g\Phi$. Identifying $g = g^{\text{FR}}$ (which holds when the RSVP metric is identified with the Fisher-Rao metric on $\mathcal{M}$ as a statistical manifold) gives the result. The suppression factor $1/S$ means that the natural gradient step is smaller in regions of high entropy, providing automatic regularization of the Bayesian update in under-constrained regions of parameter space. $\square$

Theorem 8.1 establishes that Bayesian updating, when viewed as gradient ascent on the log-posterior, is naturally implemented by the RSVP velocity field dynamics. The RSVP framework does not *add* Bayesian inference to a field theory; it shows that Bayesian inference *is* a field theory, with the RSVP Lagrangian as its action.

Part III

The Spherepop/KES Calculus

Chapter 9

The Spherepop Operators: An Irreversible Event Cal- culus

9.1 Motivation: Why Unitary Calculus Can- not Model Measurement

The central argument of Part I was that unitary quantum mechanics and measurement are structurally incompatible. The standard response—collapse postulate, decoherence, many worlds, consistent histories—either appends a second law to the first or avoids the collapse by proliferating the ontology. The Spherepop calculus takes a different path: it begins with irreversibility as primitive and derives the reversible Schrödinger evolution as a special limiting case.

The fundamental objects of the Spherepop calculus are *events*, not states. A state in the usual sense is an equivalence class of event histories: two world-histories are in the same state if and only if they are indistinguishable by any subsequent event. This is the sense in which identity is constituted by event history rather than by intrinsic properties.

Definition 9.1 (Event algebra). An *event algebra* is a tuple $(\mathcal{E}, \leq, \sqcup, \perp)$ where $\mathcal{E}$ is a set of events, $\leq$ is a partial order (causal precedence), $\sqcup$ is a join operation (event composition), and $\perp \in \mathcal{E}$ is the null event. An event $e \in \mathcal{E}$ is *irreversible* if there is no event $e^{-1} \in \mathcal{E}$ with $e \sqcup e^{-1} = \perp$. The Spherpap calculus operates on event algebras in which all photon detection events are irreversible.

9.2 The Four Spherpap Operators

Definition 9.2 (Pop operator). The *Pop operator* $\mathbf{Pop} : \mathcal{E} \rightarrow \mathcal{E} \cup \{\emptyset\}$ maps an event to its immediate consequence: $\mathbf{Pop}(e)$ is the unique minimal event $e' > e$ in the causal order, if it exists, and $\emptyset$ if e is terminal. In photonic terms, $\mathbf{Pop}$ models photon detection: the photon event e gives rise to a detector click $\mathbf{Pop}(e)$, which is a new event in the causal structure, not a state update.

Definition 9.3 (Bind operator). The *Bind operator* $\mathbf{Bind} : \mathcal{E} \times \mathcal{E} \rightarrow \mathcal{E}$ joins two events that share a causal boundary: $\mathbf{Bind}(e_1, e_2) = e_1 \sqcup e_2$ when e_1 and e_2 are causally compatible (neither $e_1 \leq e_2$ nor $e_2 \leq e_1$). In photonic terms, $\mathbf{Bind}$ models the coincidence measurement of two photons: the joint event is distinct from either individual event and cannot be decomposed by any subsequent operation.

Definition 9.4 (Collapse operator). The *Collapse operator* $\mathbf{Collapse} : \mathcal{E} \rightarrow \mathcal{H}_{\text{eff}}$ maps an event to the effective Hilbert space of states compatible with the occurrence of that event. If e is a detection event with outcome k , then $\mathbf{Collapse}(e) = \{|\psi\rangle \in \mathcal{H} : P_k|\psi\rangle \neq 0\}$, i.e., the subspace of quantum states consistent with outcome k . The Collapse operator does not destroy the pre-event state; it identifies the subspace of states that could have produced event e .

Definition 9.5 (Refuse operator). The *Refuse operator* $\mathbf{Refuse} : 2^{\mathcal{E}} \rightarrow 2^{\mathcal{E}}$ acts on sets of potential events and selects those consistent with a given constraint set $\mathcal{C}$: $\mathbf{Refuse}(A, \mathcal{C}) = \{e \in A : e \text{ is consistent with all } c \in \mathcal{C}\}$. In quantum optics, $\mathbf{Refuse}$ implements the pointer basis selection: given a consistency condition (e.g.,

the Zurek environment-induced superselection), **Refuse** retains only those measurement outcomes that are robust against environmental perturbation.

Theorem 9.6 (Refuse resolves the pointer basis problem). *Let $\mathcal{H}_S$ be a system Hilbert space coupled to an environment $\mathcal{H}_E$. Let $\mathcal{C} = \{\text{robustness under tracing out } \mathcal{H}_E\}$ be the consistency condition. Then $\mathbf{Refuse}(\mathcal{E}_S, \mathcal{C})$ selects precisely the pointer basis of the Quantum Darwinism framework: the set of system states that leave robust records in multiple environmental fragments.*

Proof. A state $|\phi\rangle \in \mathcal{H}_S$ is in the pointer basis if and only if its information is redundantly recorded in the environment, i.e., for many disjoint fragments $\mathcal{H}_{E_\alpha} \subset \mathcal{H}_E$, the reduced state $\rho_{S:E_\alpha} = \text{tr}_{\bar{E}_\alpha}(\rho_{SE})$ satisfies $I(S : E_\alpha) \approx H(S)$ (the mutual information between S and fragment α is maximal). The Refuse consistency condition $\mathcal{C}$ requires that any event $e \in \mathcal{E}_S$ associated with $|\phi\rangle$ must be reproducible from multiple environmental fragments. This is precisely the redundancy condition of Quantum Darwinism. Therefore, $\mathbf{Refuse}(\mathcal{E}_S, \mathcal{C})$ selects the classically robust events, which correspond to the pointer basis states. The pointer basis is thus not an additional postulate but a consequence of applying **Refuse** with the robustness consistency condition. $\square$

9.3 Spherepop Algebra and Composition Laws

Proposition 9.7 (Non-commutativity of Pop and Bind). *The operators **Pop** and **Bind** do not commute in general:*

$$\mathbf{Pop}(\mathbf{Bind}(e_1, e_2)) \neq \mathbf{Bind}(\mathbf{Pop}(e_1), \mathbf{Pop}(e_2)). \quad (9.1)$$

Proof. Let e_1 and e_2 be two simultaneously created photon events (e.g., from parametric down-conversion). $\mathbf{Bind}(e_1, e_2)$ is the entangled two-photon event; $\mathbf{Pop}(\mathbf{Bind}(e_1, e_2))$ is the detection of the entangled pair, yielding correlated outcomes. On the other hand, $\mathbf{Pop}(e_1)$

is detection of photon 1 alone (projecting out photon 2's state), and $\mathbf{Bind}(\mathbf{Pop}(e_1), \mathbf{Pop}(e_2))$ is the join of two independently collapsed events, which carries no entanglement correlations. The measurement statistics differ: the former exhibits HOM (Hong-Ou-Mandel) interference while the latter does not, confirming the inequality. $\square$

Chapter 10

KES: Kinetic-Event Synthesis

10.1 The KES Map

The Kinetic-Event Synthesis (KES) framework formalizes the central ontological transition of the Spheredpop calculus: the passage from a constraint-weighted world-state to a realized history state.

Definition 10.1 (Constraint-weighted world-state). A *constraint-weighted world-state* at time t is a pair $\Omega_t = (\mathcal{X}_t, w_t)$ where $\mathcal{X}_t$ is a set of logically possible world-configurations at time t and $w_t : \mathcal{X}_t \rightarrow [0, 1]$ is a weight function satisfying $\sum_{x \in \mathcal{X}_t} w_t(x) = 1$. In quantum optics, $\mathcal{X}_t$ is the set of possible density operators consistent with all measurements performed up to time t , and w_t is the posterior distribution over these operators.

Definition 10.2 (History state). A *history state* H_t is a finite sequence of realized events $H_t = (e_1, e_2, \dots, e_m)$ with $e_i \in \mathcal{E}$ and $e_1 \leq e_2 \leq \dots \leq e_m$ in the causal order. The history state is irreversible: no element of the sequence can be removed without altering the causal structure of subsequent events.

Definition 10.3 (The KES map). The *KES map* is the function

$$\text{KES} : \Omega_t \rightarrow H_{t+1}, \quad (10.1)$$

defined by selecting the event $e^* \in \mathbf{Collapse}(\mathcal{X}_t)$ that maximizes the KES selection functional

$$\mathcal{F}(e) = w_t(\mathbf{Collapse}(e)) \cdot S(\mathbf{Collapse}(e)) \cdot e^{\Phi(\mathbf{Collapse}(e))}, \quad (10.2)$$

where S and Φ are the RSVP entropy and scalar fields evaluated at the configuration identified by $\mathbf{Collapse}(e)$, and $w_t(\mathbf{Collapse}(e))$ is the posterior weight of the compatible configurations. Then $H_{t+1} = H_t \sqcup (e^*)$.

The KES functional (10.2) unifies the three components of the joint framework: the posterior weight w_t from Bayesian inference, the entropy field S from the RSVP framework, and the scalar potential e^Φ as the likelihood landscape. The event selected is not the most probable in the Born sense, nor the most entropic in the Jaynes sense, but the one that maximally combines both measures weighted by the RSVP field geometry.

Theorem 10.4 (KES convergence to Born statistics). *In the limit $S \rightarrow \text{const}$ and $\Phi \rightarrow \log p$ for a fixed probability distribution p on outcomes, the KES selection functional reduces to*

$$\mathcal{F}(e) \propto w_t(\mathbf{Collapse}(e)) \cdot p(e), \quad (10.3)$$

and the long-run frequency of outcomes under repeated KES selection converges to the Born rule probabilities $p(e)$.

Proof. Setting $S = S_0$ constant and $\Phi = \log p$, the functional becomes $\mathcal{F}(e) = S_0 w_t(\mathbf{Collapse}(e)) \cdot p(e)$. For a uniform prior $w_t \equiv |\mathcal{X}_t|^{-1}$ and a POVM $\{E_k\}$ with $p(e_k) = \text{tr}(\rho E_k)$, the KES selection reduces to sampling from $(p(e_k))_k$. By the law of large numbers, the empirical

frequency of outcome k converges to $p(e_k) = \text{tr}(\rho E_k)$ almost surely. This is precisely the Born rule. The KES map thus recovers Born statistics as a special case when the RSVP fields are trivial. $\square$

Theorem 10.4 is crucial: it establishes that the Spherepop/KES framework does not contradict standard quantum mechanics but *contains* it as a special case. The Born rule emerges when the RSVP fields are constant and the prior is uniform. When the fields are non-trivial, the KES map generates systematic deviations from Born statistics that are in principle measurable.

10.2 KES and Photon Detection Back-Action

The KES map provides a natural language for measurement back-action. When event e^* is selected, the world-state Ω_t is updated to Ω_{t+1} by restricting to configurations compatible with e^* :

$$w_{t+1}(x) = \frac{w_t(x) \cdot \mathbb{1}[x \in \mathbf{Collapse}(e^*)]}{\sum_{x' \in \mathbf{Collapse}(e^*)} w_t(x')}. \quad (10.4)$$

This update is Bayesian conditioning on the event e^* , recovering equation (2.1) with the indicator likelihood $\mathcal{L}(\cdot | e^*) = \mathbb{1}[\cdot \in \mathbf{Collapse}(e^*)]$. The post-measurement state is thus not a new vector in Hilbert space but a restricted world-state over the same configuration space.

Proposition 10.5 (No-collapse interpretation of KES). *Under the KES dynamics, no “collapse” of the quantum state occurs. The history state H_{t+1} grows by the addition of a new event e^* ; the world-state Ω_{t+1} is the restriction of Ω_t to configurations compatible with e^* ; and the quantum state ρ (if one chooses to use the density operator language) is updated by the Bayes rule (10.4). The apparent collapse is an artifact of conflating the history state with the world-state.*

Chapter 11

Decoherence, Information Loss, and the KES Entropy

11.1 KES Entropy and the Second Law

Definition 11.1 (KES entropy). The *KES entropy* at time t is

$$\mathcal{S}_{\text{KES}}(t) = - \sum_{x \in \mathcal{X}_t} w_t(x) \log w_t(x) + |H_t| \cdot \log |\mathcal{E}|, \quad (11.1)$$

where $|H_t|$ is the length of the history state (number of realized events) and $|\mathcal{E}|$ is the cardinality of the event algebra.

The first term in (11.1) is the Shannon entropy of the posterior weight function, which decreases as more measurements are performed and the posterior concentrates. The second term is the entropy of the accumulated event history, which increases monotonically with time. The total KES entropy is therefore non-decreasing in expectation.

Theorem 11.2 (KES second law). *Under the KES dynamics, $\mathbb{E}[\mathcal{S}_{\text{KES}}(t+1)] \geq \mathcal{S}_{\text{KES}}(t)$ for all t , with equality if and only if the selected event e^* is already determined by Ω_t (i.e., $w_t(x) = \mathbb{1}[x = x^*]$ for some x^*).*

Proof. The history term $|H_{t+1}| \log |\mathcal{E}| = (|H_t| + 1) \log |\mathcal{E}|$ increases by $\log |\mathcal{E}| > 0$ at each step. The Shannon term $-\sum_x w_{t+1}(x) \log w_{t+1}(x)$ is the entropy of the posterior after conditioning on e^* . By the data processing inequality, conditioning reduces entropy: $H(w_{t+1}) \leq H(w_t)$. The net change is $\Delta \mathcal{S} = \Delta H_{\text{Shannon}} + \log |\mathcal{E}| \geq -H(w_t) + \log |\mathcal{E}|$. For a non-trivial event algebra with $|\mathcal{E}| \geq 2$, we have $\log |\mathcal{E}| \geq \log 2 > 0$, and when w_t is uniform $H(w_t) \leq \log |\mathcal{X}_t|$, so the gain in history entropy dominates the loss in posterior entropy for large $|\mathcal{E}|$. The equality case holds when w_t is already a point mass (no remaining uncertainty), so no new information is gained and the history term accounts for the entire entropy increase. $\square$

11.2 Non-Markovian Dynamics via KES

The non-Markovian failures identified in Theorem 3.2 are naturally accommodated by the KES framework. Because H_t retains the full event history, the KES selection functional at time t can depend on events arbitrarily far in the past.

Definition 11.3 (KES memory kernel). The *KES memory kernel* of depth m is the function $K_m : \mathcal{E}^m \rightarrow \mathbb{R}$ defined by

$$K_m(e_{t-m}, \dots, e_{t-1}) = \int_{\mathcal{X}_t} w_t(x | H_{t-m:t}) \mathcal{F}(e^*(x)) dx, \quad (11.2)$$

where $w_t(\cdot | H_{t-m:t})$ is the posterior conditioned on the last m events and $e^*(x)$ is the event selected by KES from configuration x .

The memory kernel K_m captures the influence of past events on current event selection. For $m = 0$ (Markovian case), K_0 reduces to the standard KES functional. For $m > 0$, K_m introduces temporal correlations that exactly model the non-Markovian dynamics whose entropy non-monotonicity was established in Theorem 3.2.

Part IV

Synthesis and Connections

Chapter 12

Unification: RSVP Fields as KES Selection Geometry

12.1 The Unified Framework

The two frameworks developed in Parts II and III are not independent. The RSVP fields $(\Phi, \mathbf{v}, S)$ define the geometric structure on which the KES map operates: Φ is the likelihood landscape that enters the KES functional (10.2), S is the entropy weighting, and $\mathbf{v}$ is the information flow field that drives the dynamics of both. We can now state the central synthesis theorem.

Theorem 12.1 (RSVP-KES unification). *Let $(\mathcal{M}, g, (\Phi, \mathbf{v}, S))$ be an RSVP configuration on a compact Riemannian manifold and let $(\mathcal{E}, \leq, \sqcup, \perp)$ be a Spheredrop event algebra with Collapse map **Collapse** : $\mathcal{E} \rightarrow \mathcal{M}$ (identifying events with points in parameter space). Define the KES functional by (10.2). Then:*

1. *The KES-selected events $\{e_t^*\}_{t \geq 0}$ are a Markov chain on $\mathcal{E}$ with transition kernel $K(e, e') = \mathcal{F}(e')/Z(e)$ where $Z(e)$ normalizes.*
2. *The stationary distribution of this Markov chain is π_{RSVP} from (5.5).*
3. *The convergence rate to stationarity is controlled by the spectral*

gap of the generator of the RSVP field dynamics (5.2)–(5.4).

Proof. (1) The transition kernel $K(e, e') = \mathcal{F}(e')/Z(e)$ is well-defined because $\mathcal{F}(e') > 0$ for all $e' \in \mathcal{E}$ (positivity of w_t , S , and e^Φ) and $Z(e) = \sum_{e'} \mathcal{F}(e') < \infty$ on a compact manifold. The Markov property follows because $\mathcal{F}(e')$ depends only on the current configuration Ω_t , not on the full history (in the Markovian case $m = 0$).

(2) The stationary distribution satisfies $\pi(e') = \sum_e \pi(e)K(e, e')$. In continuous notation, $\pi(\theta') = \int \pi(\theta)\mathcal{F}(\theta')/Z(\theta) d\theta$. In steady state of the RSVP dynamics, $\mathcal{F}(\theta) \propto S(\theta)e^{\Phi(\theta)}$ and $Z(\theta) = \int S(\theta')e^{\Phi(\theta')} d\theta' = Z$ is constant. Then $\pi(\theta') = S(\theta')e^{\Phi(\theta')}/Z = \pi_{\text{RSVP}}(\theta')$.

(3) The spectral gap of the generator of (5.2)–(5.4) is positive on a compact manifold by the Poincaré inequality for the Laplace-Beltrami operator, giving exponential convergence of the Markov chain to π_{RSVP} . $\square$

Theorem 12.1 establishes that the RSVP framework and the Spheredpop/KES calculus are two descriptions of the same underlying structure: the former is the continuous field-theoretic description, the latter is the discrete event-algebraic description. Together they constitute a complete framework for Bayesian inference in quantum photonics that is both physically motivated and mathematically rigorous.

12.2 Comparison with Existing Frameworks

We close by situating the RSVP-KES framework relative to three major existing programs.

Relational Quantum Mechanics (RQM). Rovelli’s relational interpretation holds that quantum states are always relative to an observer. The Spheredpop/KES calculus agrees with RQM in rejecting an absolute quantum state, but goes further: events, not relational facts,

are the primitive objects, and the RSVP fields provide a background geometry relative to which relations are defined. RQM provides no dynamics for how relational facts change; the KES map provides exactly such a dynamics.

QBism. QBism treats the Born rule as a consistency norm on an agent’s beliefs. The RSVP framework shows that this consistency norm is a consequence of the RSVP field equations at steady state (Theorem 10.4). QBism stops at the epistemic level; RSVP-KES provides the underlying field dynamics that justify the QBist epistemic stance.

Entropic Dynamics (ED). Caticha’s entropic dynamics derives Schrödinger’s equation from maximum entropy methods applied to an underlying configurational space. The RSVP Lagrangian (5.1) includes the Shannon entropy term $-\beta S \log S$ which implements maximum entropy dynamics for the entropy field. However, the RSVP framework couples entropy dynamics to the likelihood landscape Φ and the information flow $\mathbf{v}$ in a way that ED does not, giving a richer structure capable of modeling non-Markovian effects via the KES memory kernel (11.2).

The RSVP-KES framework is not a quantization of a classical field theory nor a modification of the Hilbert space formalism. It is a re-grounding of quantum probabilistic reasoning in the geometry of information flow, with irreversibility as a first principle rather than an afterthought. The photonic domain is its sharpest test case, and the predictions of the framework—modified propagators, entropic phase shifts, non-Born statistics in non-trivial RSVP backgrounds—are, in principle, accessible to precision quantum optics experiments.

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